Marwa Salah

Remote Telecommunication Engineer

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SUMMARY

Telecommunication Engineer with strong expertise in wireless (2G/3G/4G/5G), optical, and microwave transmission systems. Specialized in RF planning, network design using Atoll, and advanced optical solutions (DWDM, OTN). Proven experience in end-to-end telecom project design and remote coordination. Adept at troubleshooting, documentation, and team collaboration. Passionate about delivering optimized and innovative network solutions while working remotely.

EXPERIENCE

Wireless Transmission Engine

My Communication
Sep 2023 – Sep 2024
| Jordan

- Designed and implemented BTS sites for 2G, 3G, and 4G networks with efficient RF planning using Atoll.
- Conducted detailed coverage and capacity analysis for GSM, UMTS, and LTE.
- Handled site surveys, equipment selection, installation plans, and system integration.
- Worked with multi-vendor equipment (Huawei, Nokia, Ericsson).
- Contributed to 5G planning, coverage analysis, URLLC/mMTC/eMBB use cases, and antenna placement strategies.
-Have experience in Site selection, Survey, Site acquisition, Design (2G.3G.4G.5G power calculation), QoS, Materials configuration, Installation, Integration, optimization and remote handover.

Optical Transmission Engineer

My Communication
9/2024 - 2/2025
Jordan
I specialized in High-Level Design (HLD), Low-Level Design (LLD), Technical Design, and Bill of Quantities (BOQ), with expertise in OSN 9800, OSN 8800, and OSN 1800. Experienced in installation, handover processes, and ensuring seamless integration of optical transmission systems for optimal performance. Proficient in ISP Architecture of Optical Access Network, PON, WDM, CWDM/DWDM, OLT, Optical amplifiers, and OTN. Skilled in OSP definitions, systems, components, splicing, and testing. Designed and implemented a high-speed fiber optic network spanning 40km and 20km using G.652 fiber, with a fiber loss strategy of 0.2 dB/km and a 5dB margin for signal integrity. Implemented protection schemes including ponder on the client side and OLP/DCP, M520SM on the line side

Installed and configured two 9800 M12 devices, one for the east and one for the west, with separate racks for each device. Engineered for a capacity of 2x100GE and a line rate of 200G, with a channel scalability plan from 40 to 96 channels. Designed and implemented a ROADM system for CDC

Planned network architecture, selected suitable equipment, optimized transmission paths, and troubleshooted and maintained optical transmission systems. Completed within budget, including 10% spare parts and OPEX for 5 years

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Microwave Transmission Engineer

My Communication

02/2025 – present`

Jordan

- Designed and implemented microwave links for telecom networks, conducting site surveys, frequency planning, and equipment configuration using Huawei RTN910 and RTN950 for 2G, 3G, and LTE networks
- Analyzed Line of Sight (LOS) availability for new links using Google Earth and Pathloss, arranging surveys for verification. Completed Pathloss Engineering Design and prepared Link Budgets for customer approval and implementation in 2G, 3G, and LTE networks
- Expert in: Link type Propagation, Polarization, Transmission signal, Antenna Gain, Half power 3db, Cross polarization, Front to back ratio (Frequency, Band, Wave length, Distance. LOS, nLOS, NLOS) Definition, Calculation and types MW Design, topology,

network Architecture, IDU, ODU, Xpic Protection, Diversity (FD, SD), N+N Link, MIMO, sectorization, Multi bands ODU PDH, SDH (Protocol) And Frame, Modulation, Alignment.

TRAINING / COURSES

- Integrated Communication Engineering from My Communication Academy.
- CCNA.
- IT

SKILLS

-(DWDM, OTN, 5G Planning).
- Remote Collaboration (Zoom, Google Meet)
- Microsoft Office (Word, Excel)
- Report Writing & Technical Documentation
- (Team Collaboration, Problem Solving).

REFERENCE

Available upon request

EDUCATION

Bachelor's degree in Telecommunication Engineering

Suez Canal University

(B.S.C) in Electrical Engineering (Electronics and Communications Engineering).

Graduation year: 2007.